[['MMCATrack: Multi-Modal Channel Attention Tracker',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cvi2.70060?af=R',

'We proposes a multi-modal channel attention tracking algorithm wherein a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features.\n\n\n\n\n\n\nABSTRACT\nMost existing Transformer-based visual object tracking methods rely exclusively on the feature map from the last encoder layer for object prediction, thereby overlooking the rich information contained in shallow and intermediate layer feature maps. This limitation reduces the representational capacity of the model. Moreover, current multi-modal tracking frameworks typically construct multi-modal features through simple concatenation, which fails to adequately account for the differential contributions of individual modalities to the final prediction task. As a result, these approaches exhibit an insufficient ability to express key features within the multi-modal representation. To address the aforementioned issues, this paper proposes a multi-modal channel attention tracking algorithm, where a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features. Specifically, the multi-modal channel attention block first aggregates multi-modal information from the multi-layer feature maps of the encoder through cross layer cascading and then applies channel attention mechanism to dynamically calibrate the channel weights in the generated multi-modal features, thereby enhancing the representation of key features. In addition, this article proposes a new regression loss function to improve localisation accuracy. Finally, abundant experiments conducted on five benchmarks including GOT-10K, TrackingNet, TNL2K, VisEvent and RGBT234 have verified the effectiveness of our theory.',

'Zhiqiang Zhao, Daitu Wen, Yuanhang Gu, Xiaoli Luo, Tao Ma, Xu Ma, Bin Wu',

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['PhantomVeil: Physical Camouflage for Multi-Object Tracking Deep Neural Networks',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70353?af=R',

'PhantomVeil, a physical camouflage attack method for multi-object tracking, systematically exploits the structural vulnerabilities in the association between detection and tracking modules, thereby constructing a physically realizable adversarial generation framework.\n\n\n\n\n\n\n\n\nABSTRACT\nMulti-object tracking (MOT) models have achieved great success in wide ranging applications, such as video surveillance and autonomous driving, especially for those deep neural networks (DNNs) based MOT, which obtained the state-of-the-art detection performance. Unfortunately, existing research has unveiled that DNNs are vulnerable to carefully crafted adversarial perturbations, leading to catastrophic consequences. In order to explore the adversarial vulnerability of MOT, we propose PhantomVeil, a novel physical camouflage for DNN based MOT. PhantomVeil significantly differs from previous work in four aspects: (1) effectiveness—by undermining the data association between the detection and tracking stages of MOT models, our method reliably ensures a high probability of attack success; (2) generality—by targeting the common core mechanisms of MOT models, we design an efficient loss optimization strategy that enables generalized attacks across different models; (3) transferability—Gaussian smoothing and color gamut constraints are introduced, it reduces the adversarial attenuation when transferring from the digital domain to the physical domain and improves the effectiveness of target transfer in the physical domain by restricting drastic pixel value fluctuations and ensuring printable color values; (4)

practicability—extensive experiments conducted on two public datasets and one self-built dataset demonstrate that PhantomVeil outperforms four baseline methods on three multi-object trackers (with digital-domain attack performance reaching ×$\\times$1.02 ~$\\sim$ 6.07 that of the baselines and physical-domain attack performance reaching ×2.19~$\\times 2.19\\sim$6.07 that of the baselines), verifying its advantages in practicality, effectiveness, and\xa0generalization.',

'Jinyin Chen, Huanran Li, Ye Han',

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[['Enhancing feature representation in Siamese networks for object tracking with ranking-based loss',

'https://www.sciencedirect.com/science/article/pii/S1077314226001529',

'',

'Sachin Sakthi K.S., Young Hoon Joo, Jae Hoon Jeong',

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'2026',

'5'],

['Global–local distillation network-based audio–visual speaker tracking with incomplete modalities',

'https://www.sciencedirect.com/science/article/pii/S0031320326008812',

'',

'Yidi Li, Yihan Li, Zhenhuan Xu, Weiwei Wan, Hong Liu',

'2',

'2026',

'11'],

['Evolving Markov Chains: Online Mode Discovery and Recognition From Data Streams',

'http://ieeexplore.ieee.org/document/11408039',

'Markov chains are simple yet powerful mathematical structures to model temporally dependent processes. They generally assume stationary data, i.e., fixed transition probabilities between observations/states. However, live, real-world processes, like in the context of activity tracking, biological time series, or industrial monitoring, often switch behavior over time. Such behavior switches can be modeled as transitions between higher-level modes (e.g., running, walking, etc.). Yet all modes are usually not previously known, often exhibit vastly differing transition probabilities, and can switch unpredictably. Thus, to track behavior changes of live, real-world processes, this study proposes an online and efficient method to construct Evolving Markov chains (EMCs). EMCs adaptively track transition probabilities, automatically discover modes, and detect mode switches in an online manner. In contrast to previous work, EMCs are of arbitrary order, the proposed update scheme does not rely on tracking windows, only updates the relevant region of the probability tensor, and enjoys geometric convergence of the expected estimates. Our evaluation of synthetic data and real-world applications on human activity recognition, electric motor condition monitoring, and eye-state recognition from electroencephalography (EEG) measurements illustrates the versatility of the approach and points to the potential of EMCs to efficiently track, model, and understand live, real-world processes.',

'Kutalmış Coşkun, Borahan Tümer, Bjarne C. Hiller, Martin Becker',

'3',

'2026',

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['Real-CD: Change Detection Under Real-World Complex Interference via Dynamic Distribution Correction',

'http://ieeexplore.ieee.org/document/11489274',

'While change detection (CD) is crucial for tracking dynamic changes on the Earth’s surface, it faces substantial challenges in real-world settings caused by seasonal variations and sensor-related interference. Current CD models often suffer performance degradation under such conditions, mainly due to two key challenges. First, most existing CD datasets lack sufficient temporal and environmental diversity, as they are typically collected over constrained time spans. This limits the models’ ability to generalize across varying conditions. Second, many CD methods are heavily data-driven and rely on simplified assumptions, leading to models that are not adequately designed to handle the complex, heterogeneous nature of real-world scenarios. Together, these challenges restrict the robustness and practical applicability of current CD approaches. To overcome these challenges, we make the following contributions in this paper: 1) Regarding data diversity, we construct a comprehensive benchmark by introducing five typical perturbations (fog, snow, motion blur, Gaussian noise, and impulse noise) into three classical CD datasets and supplementing them with a real-world seasonal dataset, resulting in 75 interference-rich scenarios. This enables a systematic evaluation under diverse real-world conditions, revealing that such perturbations induce severe distribution shifts across both temporal phases and hierarchical network layers, leading to substantial performance degradation in existing models. 2) Algorithmically, we propose Real-CD, a novel method specifically designed to address distribution shifts in real-world CD. The core of Real-CD is to leverage bi-temporal correlations to perform adaptive distribution alignment across hierarchical layers and temporal phases. Specifically, we propose the Distribution Shifts Alleviation Module (DSAM) to correct distribution shifts. The DSAM captures bi-temporal differences and similarities to formulate temporal-specific adjustment strategies for each LayerNorm (LN) layer. To stabilize the optimization of DSAM, we propose the Distribution Consistency Optimization Strategy (DCOS), which introduces a flip-based auxiliary task that encourages the model to maintain distributional consistency under complex bi-temporal disturbances. Consequently, our method outperforms other state-of-the-art approaches and achieves the best performance on the proposed dataset. Our datasets and code implementation will be available at https://github.com/fangyee-ISALAB/Real-CD',

'Leyuan Fang, Yi Fang, Pedram Ghamisi, Qiang Liu',

'4',

'2026',

'5'],

['Fuzzy Logic-Enhanced Neuroadaptive Fault-Tolerant Control for Vehicular Platoons With Stochastic Disturbances and Asymmetric Spacing Constraints',

'http://ieeexplore.ieee.org/document/11259115',

'This article introduces a novel fuzzy logic-enhanced neuroadaptive sliding mode control (FLENNSMC) framework, developed for vehicular platoon systems subject to a confluence of challenges. Leveraging the synergistic integration of fuzzy logic's interpretive strengths and neural networks' adaptive learning capabilities, FLENNSMC effectively addresses nonlinear dynamics, stochastic disturbances, actuator faults, and stringent asymmetric spacing constraints. We propose a Takagi–Sugeno (T-S) fuzzy model to structure the learning process and a fuzzy logic-enhanced RBFNN (FLERBFNN) for robust approximation of unknown functions, including unmodeled dynamics and fault signals. The controller design incorporates a fault-tolerant control mechanism for enhanced robustness, an asymmetric barrier Lyapunov function (BLF) to strictly enforce spacing constraints, and a Nussbaum function to compensate for actuator faults with unknown directions. The fuzzy logic-enhanced structure allows for localized and efficient learning, which reduces computational burden and improves adaptation speed. Through a rigorous stochastic Lyapunov–Krasovskii stability analysis, we derive sufficient LMI-based conditions for the uniform ultimate boundedness (UUB) of tracking errors in the mean square sense and guarantee a mixed H-infinity/passivity performance. Extensive simulations on a 2-D multilane vehicular platoon demonstrate the superior performance of the proposed FLENNSFC compared to conventional neuroadaptive control approaches, particularly highlighting the benefits of fuzzy logic in structuring the learning process and handling complex uncertainties. Simulation code is available at https://github.com/zhanganguo/FLENNSMC-Platoon-Control-Simulation',

'Yao Wen, Xiaohong Chen, Xuesong Xu, Anguo Zhang, Yongfu Li',

'5',

'2026',

'5'],

['Adaptive Optimal Prescribed Performance Tracking Solutions for Multiplayer Systems With Error Constraints',

'http://ieeexplore.ieee.org/document/11259067',

'This brief presents the adaptive optimal prescribed performance tracking solutions for the multiplayer nonlinear systems based on the adaptive critic learning scheme, where the tracking errors are constrained to a predefined bounded set. First, the general optimal tracking solutions of multiplayer nonlinear systems are presented. Every optimal tracking solution of multiple players consists of the steady-state part and the adaptive feedback part. The steady-state part can be obtained directly according to the tracking signal and system dynamics. Then, the adaptive feedback part can be studied with the prescribed performance constraints and adaptive critic learning such that multiple value functions achieve a Nash equilibrium with error constraints. Moreover, the convergence of the critic network weight is analyzed by the Lyapunov algorithm. Finally, simulation results and experiments are presented to demonstrate the satisfactory performance of the proposed method.',

'Yongfeng Lv, Huimin Chang, Jun Zhao, Yuteng Tian',

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'2026',

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['MMCATrack: Multi-Modal Channel Attention Tracker',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cvi2.70060?af=R',

'We proposes a multi-modal channel attention tracking algorithm wherein a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features.\n\n\n\n\n\n\nABSTRACT\nMost existing Transformer-based visual object tracking methods rely exclusively on the feature map from the last encoder layer for object prediction, thereby overlooking the rich information contained in shallow and intermediate layer feature maps. This limitation reduces the representational capacity of the model. Moreover, current multi-modal tracking frameworks typically construct multi-modal features through simple concatenation, which fails to adequately account for the differential contributions of individual modalities to the final prediction task. As a result, these approaches exhibit an insufficient ability to express key features within the multi-modal representation. To address the aforementioned issues, this paper proposes a multi-modal channel attention tracking algorithm, where a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features. Specifically, the multi-modal channel attention block first aggregates multi-modal information from the multi-layer feature maps of the encoder through cross layer cascading and then applies channel attention mechanism to dynamically calibrate the channel weights in the generated multi-modal features, thereby enhancing the representation of key features. In addition, this article proposes a new regression loss function to improve localisation accuracy. Finally, abundant experiments conducted on five benchmarks including GOT-10K, TrackingNet, TNL2K, VisEvent and RGBT234 have verified the effectiveness of our theory.',

'Zhiqiang Zhao, Daitu Wen, Yuanhang Gu, Xiaoli Luo, Tao Ma, Xu Ma, Bin Wu',

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'2026',

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['Measurement for Rock Bolt Pitch and Row Pitch Based on Dual-Modal Image Analysis',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70381?af=R',

'A measurement algorithm of rock bolt pitch and row pitch based on deep learning and computer vision is proposed. Deep learning enables the localisation of rock bolts in images, while computer vision enables spatial measurements in images, which are well suited to the task of measuring the pitch and row pitch of rock bolts. \n\n\n\n\n\n\n\nABSTRACT\nThe pitch and row pitch of rock bolts are critical parameters in rock-support techniques. These parameters facilitate tracking construction processes and monitoring bolt displacement for early warning of rock bursts and roof falls. A measurement method for these pitches based on dual-modal image recognition is proposed to solve the problems of high personnel safety risks, low accuracy, and high labour intensity in traditional measurement. First, an intrinsically safe structured-light camera captures colour images and point clouds, and Gaussian heatmap-based point annotation is conducted on the colour images to produce training samples. Second, the detection network is enhanced via point feature maps, feature point attention, and mixed up-sampling. A weighted Gaussian heatmap is adopted as the loss function to develop a rock bolt point detection network, which effectively enhances the detection and localisation accuracy. Finally, a two-stage serial fitting method is proposed to solve the linear equation for each row and complete the measurement. Simulation experiments demonstrate that the developed point detection network achieves notable gains in precision, recall, and positioning error. The precision is 87.53%, the recall is 93.03%, and the positioning error is 11.33\xa0mm, all ranking among the top performers. Industrial experiments demonstrate that the method is well-suited for on-site applications.',

'You Lei, Yang Xiaoyu, Luo Chuan, Li Tao, Zhang Haifeng, Liu Yanchi',

'7',

'2026',

'5'],

['YOLOViT-DETECT: A Hybrid Approach for Real-Time Fish Species Detection and Classification in Aquaculture',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70373?af=R',

"We propose YOLOViT-DETECT, a hybrid model combining an enhanced YOLOv8 with a spatial attention module for fish detection and a ViT-B/16 transformer for species classification. The method achieves high accuracy in challenging underwater environments, outperforming standalone models. This approach enables real-time, reliable fish monitoring in aquaculture.\n\n\n\n\n\n\n\n\nABSTRACT\nThe ability to differentiate between different species of fish plays an essential role in aquaculture, as it helps to protect their populations and monitor their welfare. However, conventional learning methods are performance-limited, especially under complex submarine conditions such as similarity of fish body and background, low light levels and poor water quality. This paper proposes a new method titled YOLOViT-DETECT, which has practical significance for aquaculture monitoring by enabling accurate, automated tracking of fish species, supporting fish welfare, efficient farm management, and sustainable production. The approach is distinguished by its hybrid and attentional architecture which jointly optimizes the detection and classification of fish in complex underwater conditions. YOLOViT-DETECT uses an improved YOLOv8 model for the detection phase and a ViT-B/16 model for fish classification. To reinforce the model's focus on relevant areas of the image, even in the presence of a complex background, we have added a spatial attention module (SAM) to YOLOv8. Detection results from YOLOv8 Improved and classification results from ViT were merged into a single output. Experiments have shown that YOLOViT-DETECT is a robust method whose results exceed those achieved with YOLOv8 and ViT-B/16 separately. The precision rate reached 0.9033 in the detection phase and 0.9524 in the classification\xa0phase.",

'Mahdi Hamzaoui, Mohamed Ould-Elhassen Aoueileyine, Imen Filali, Ridha Bouallegue',

'7',

'2026',

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['PhantomVeil: Physical Camouflage for Multi-Object Tracking Deep Neural Networks',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70353?af=R',

'PhantomVeil, a physical camouflage attack method for multi-object tracking, systematically exploits the structural vulnerabilities in the association between detection and tracking modules, thereby constructing a physically realizable adversarial generation framework.\n\n\n\n\n\n\n\n\nABSTRACT\nMulti-object tracking (MOT) models have achieved great success in wide ranging applications, such as video surveillance and autonomous driving, especially for those deep neural networks (DNNs) based MOT, which obtained the state-of-the-art detection performance. Unfortunately, existing research has unveiled that DNNs are vulnerable to carefully crafted adversarial perturbations, leading to catastrophic consequences. In order to explore the adversarial vulnerability of MOT, we propose PhantomVeil, a novel physical camouflage for DNN based MOT. PhantomVeil significantly differs from previous work in four aspects: (1) effectiveness—by undermining the data association between the detection and tracking stages of MOT models, our method reliably ensures a high probability of attack success; (2) generality—by targeting the common core mechanisms of MOT models, we design an efficient loss optimization strategy that enables generalized attacks across different models; (3) transferability—Gaussian smoothing and color gamut constraints are introduced, it reduces the adversarial attenuation when transferring from the digital domain to the physical domain and improves the effectiveness of target transfer in the physical domain by restricting drastic pixel value fluctuations and ensuring printable color values; (4) practicability—extensive experiments conducted on two public datasets and one self-built dataset demonstrate that PhantomVeil outperforms four baseline methods on three multi-object trackers (with digital-domain attack performance reaching ×$\\times$1.02 ~$\\sim$ 6.07 that of the baselines and physical-domain attack performance reaching ×2.19~$\\times 2.19\\sim$6.07 that of the baselines), verifying its advantages in practicality, effectiveness, and\xa0generalization.',

'Jinyin Chen, Huanran Li, Ye Han',

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'2026',

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['YOLO-O: An Improved YOLO-Based Framework for Vehicle Detection',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70337?af=R',

"This study proposes a YOLO-O model, based on YOLOv7, for improved detection of tiny vehicles on roads. Enhancements like residual networks, Squeeze and Excitation Network (SENet), coordinate attention, and SIoU regression loss boost precision, especially for small objects. The model achieved a superior mAP of 94.9% on COCO, outperforming previous YOLO versions, and showed strong performance on PASCAL Visual Object Classes and LISA datasets.\n\n\n\n\n\n\n\n\nABSTRACT\nFor road safety, human factors and ergonomics play a vital role, influencing the object detection. To automate the object detection, several algorithms like R-CNN, SSD, R-FCN, and YOLO have been used for this task. While these models are effective for many object detection applications, detecting tiny objects on roads remains challenging due to low precision. Developing a lightweight and robust model capable of high-precision detection of tiny objects is essential. This study proposes a YOLO-O model, built on an enhanced YOLOv7 model, to detect tiny vehicles. A residual network with three convolutional layers and Squeeze and Excitation Network (SENet) replaces the Efficient Layer Aggregation Network (ELAN) in YOLOv7. Coordinate attention modules are added to improve recognition accuracy, and the SIoU regression loss enhances bounding box prediction. Moreover, an 80 × 80 detection head improves the recognition of small vehicles. The model, tested on COCO, achieved superior performance with an mAP of 94.9%, outperforming YOLOV2, YOLOV3, and YOLOV5. Moreover, the proposed model performed significantly on cross-validation using the PASCAL Visual Object Classes and LISA datasets. This system demonstrates the method's practicality and robustness for tiny vehicle recognition, offering a foundation for developing automated object detection and tracking.", 'Rabbia Mahum, Usama Irshad, Ahmed M. El-Sherbeeny, Mohammad Shehab, Seongkwan Mark Lee',

'7',

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['Advancing Traffic Safety in Dense Urban Settings: A Robust Real-Time Wrong-Way Driving Detection System with Hough Transform and YOLOv5',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70322?af=R',

'The study presents a real-time wrong-way driving detection system that integrates Hough Line Transform for road boundary extraction and YOLOv5 for vehicle detection, achieving 96% accuracy and a 0.05% false positive rate. Validated in Chattogram, Bangladesh, the system demonstrates a 60.05% detection rate across diverse urban traffic scenarios using a novel dataset of 100 videos. This innovative approach enhances traffic safety in dense urban settings by addressing the scarcity of context-specific data.\n\n\n\n\n\n\n\n\nABSTRACT\nIn Bangladesh, where a burgeoning population strains limited road networks, wrong-way driving (WWD) exacerbates traffic congestion and elevates accident risks, posing a critical public safety challenge. We present a pioneering real-time WWD detection system that integrates cutting-edge image processing and deep learning to transform traffic monitoring. Our methodology combines the Hough line transform for precise road boundary extraction with the state-of-the-art YOLOv5 algorithm for high-fidelity, real-time vehicle detection. Vehicle direction is inferred from single-frame orientation (distinguishing front/rear views) without multi-frame tracking, ensuring lightweight operation and scalable deployment. Tested on a novel, region-specific dataset from Chattogram, the system achieves an impressive 96% vehicle detection accuracy (mAP@0.5) during training and validation. Real-world evaluations on 20 diverse test videos yield an overall 60.6% recall for wrong-way vehicles, with a representative subset of 10 videos reaching 75% recall (and 41% precision). This research advances computer vision applications in traffic safety, providing an effective and deployable framework for reducing WWD incidents in high congestion urban regions and paving the way for enhanced intelligent transportation systems.',

'Anik Majumder, Joya Mallick, Dhrubajyoti Das, Avishek Chowdhury, Shrabanti Debi',

'7',

'2026',

'3'],

['Robust Radar Tracking via Joint Adaptive EKF and Power Allocation Under Dynamic Clutter',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70601?af=R',

'ABSTRACT\nIn this letter, a joint adaptive extended Kalman filter (AEKF) and power allocation strategy is proposed for robust multitarget tracking in colocated MIMO radar systems under dynamic clutter. The unknown clutter intensity is incorporated into the measurement model and characterized by a truncated inverse gamma distribution. Using variational Bayesian inference, the proposed AEKF jointly estimates the target state and clutter intensity. Based on these estimates, a power allocation strategy is developed using the predicted conditional Cramér–Rao lower bound as a performance metric. Simulation results demonstrate that the proposed method achieves improved tracking performance in dynamic clutter environments.', 'Hao Sun, Yunhua Luo, Zhongjun Yu',

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['Robust Sub-Filter Based APA With Parametric Welsch Function for Acoustic Echo Cancellation Under Impulsive Noise',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70595?af=R',

'Acoustic echo cancellation (AEC) in actual communication systems is challenging due to highly correlated inputs, impulsive disturbances, and computational limitations. The traditional Affine Projection Algorithm (APA) is better than the Normalised Least Mean Square method, although it involves matrix inversion complexity and is susceptible to outliers (like impulsive noise). This paper presents a parametric Welsch-based APA with sub-filters to address these restrictions. The proposed formulation exhibits superior performance without the necessity for matrix inversion calculations. This has been demonstrated through simulation under various settings.\n\n\n\n\n\n\n\n\nABSTRACT\nThe affine projection algorithm (APA) is widely used for acoustic echo cancellation, but faces challenges like slow convergence, high steady-state errors and sensitivity to impulsive noise. In this paper, a parametric Welsch APA has been introduced and integrated into the multiple sub-filter-based framework to address these issues for highly correlated input signals and impulsive noise. The employed parametric Welsch cost function robustly suppresses impulsive noise without any matrix inversion operation as in APA. On the other hand, a multiple sub-filter framework improves convergence and tracking performance. Through a theoretical stability and convergence analysis, the performance limits have been validated. Simulations performed using speech and synthetic signals with Bernoulli–Gaussian and impulsive noise demonstrate that the proposed approach achieves superior steady-state and convergence performance than existing APA\xa0variants.',

'Gagandeep Singh, Asutosh Kar, Vasundhara, Monorama Swain, Jesper Rindom Jensen, Mads Græsbøll Christensen',

'8',

'2026',

'5'],

['An Improved Robust Filter Algorithm for Manoeuvering Target Tracking With the Unknown and Time-Varying Noise',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70548?af=R',

'The unstable radar measurement noise in natural environments degrades tracking performance. This Letter introduces a noise-adaptive matrix to efficiently compute a corrected fading factor, enabling real-time compensation for noise variations. Superior performance is verified through comparison with existing methods.\n\n\n\n\n\n\n\n\nABSTRACT\nIn the natural environment, the unstable radar measurement noise can degrade the tracking performance of the manoeuvering target. In this Letter, the iterative formulation of the noise is simplified, and the noise-adaptive matrix is introduced to calculate the fading factor of the strong tracking algorithm so that the effect of measurement noise on the fading factor can be corrected in real-time. The superior performance of the proposed method is verified by comparison with three existing improved methods on a typical example.',

'Tianhao Liu, Xi Chen, Naichang Yuan',

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['Compact Acoustic Signal Portraits for Resource-Efficient Mechanical Fault Detection',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70551?af=R',

'Dominant acoustic features are tracked to form compact signal portraits representing oscillatory modes. Faults are detected by portrait deviation, enabling low-complexity condition monitoring of moving machinery.\n\n\n\n\n\n\n\n\nABSTRACT\nA resource-efficient semi-analytical method for mechanical fault detection via acoustic signal analysis is presented. A compact signal portrait is constructed by tracking dominant salient-frequency peaks over time, without a priori assumptions on peak locations. Peak feature statistics are aggregated into a 3D matrix that encodes nominal oscillatory modes. Fault detection is achieved by measuring the deviation between current and healthy portraits using a single decision threshold. Experiments on internal combustion engine recordings demonstrate improved separability and lower computational cost compared to empirical mode decomposition, confirming suitability for resource-constrained monitoring.',

'Oleksandr Volkov, Oleksandr Bushma, Artem Sieriebriakov, Volodymyr Taranukha, Yaroslav Linder',

'8',

'2026',

'2'],

['A Novel Quantised Image Sensing for Machine Vision',

'https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70515?af=R',

'This paper shows that the image information quantised in the range of 1 to 4 bits can be utilised for object detection and tracking without any performance degradation. Thus, it is expected that the proposed technique holds promise for use in machine vision, not only maintaining the accuracy of detection and tracking, but even enabling more optimised information transfer between the image sensor and the computing system.\n\n\n\n\n\n\n\n\nABSTRACT\nThis paper proposes a quantisation-driven data-compression framework for machine vision sensors by combining nonlinear quantisation and bit-shift masking. The proposed approach compresses pixel data to an effective 1–4 bits per pixel while preserving edge-salient information required for object detection and feature tracking. Experimental results show that the compressed/quantised images maintain detection and tracking performance without degradation, indicating the proposed framework can reduce sensor-to-processor bandwidth and optimise required bit resolution.',

'Paul K. J. Park',

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